

RAM MARUTHY

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PROFESSIONAL SUMMARY

- Over 10 years of experience in machine learning, data science, and data engineering, delivering high-impact solutions across diverse sectors, including telecommunications, automotive, retail, and healthcare.
- Proficient in developing and deploying predictive models using Python, TensorFlow, Scikit-learn, and R Studio, with a focus on Neural Networks, SVM, and Decision Trees for optimizing business processes.
- Expert in managing big data environments, utilizing AWS services (S3, EC2, EMR) and Spark (Core, SQL, Streaming) to ensure scalable, real-time data processing and storage solutions.
- Advanced in Natural Language Processing (NLP), applying techniques to analyze and extract insights from customer feedback and social media data, enhancing marketing strategies and customer engagement.
- Skilled in database management, including NoSQL (MongoDB, Hive) and PL/SQL, ensuring efficient data handling and supporting complex machine learning applications.
- Experienced in automating ETL workflows using Airflow and AWS EC2, streamlining data preprocessing and enhancing operational efficiency.
- Proficient with Git and GitHub for version control, ensuring robust code management and collaboration in multi-team projects.
- Adept at deploying machine learning models on cloud platforms like AWS and integrating them with existing systems using Flask and RESTful APIs for real-time analytics.
- Skilled in data visualization using Matplotlib and Pandas, creating impactful dashboards and visual tools to support strategic decision-making.
- Experienced in leading cross-functional teams to implement AI-driven solutions, contributing to innovation and competitive advantage across business units.

SKILLS

Programming Languages: Python, C++, SQL, Javascript

Machine Learning: TensorFlow, PyTorch, Keras, scikit-learn, Theano, Caffe, MLbase, MAHOUT, MLLib, Mlflow

Gen Ai and Deep learning Models: LLM-Langchain , Llama, Transformers(BERT, Huggingface

Frameworks : Flask, Django, React

Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn

Data Integration and Reporting: SSIS, PowerBI, Tableau

Database: SQL Server, MySQL, PostgreSQL, Sqlite, MongoDB, PL/SQL

Feature Store: Hopsworks, AWS SageMaker

Big Data and Cloud: Spark core, Spark SQL, Spark streaming, HDFS, Kafka, AWS S3, EC2,EMR

Version Control: Git, Github

PROFESSIONAL EXPERIENCE

AI Chatbot Developer

Nationwide Insurance

Jan 2024 - Present

Columbus, OH

- Utilized Hugging Face Transformers to build and optimize a generative chatbot, leveraging pre-trained models and fine-tuning them to enhance conversational abilities for various use cases, including customer support and sales.
- Integrated neural networks and Support Vector Machines (SVM) within the chatbot's framework, improving the prediction of user intents and increasing the accuracy of responses, resulting in better user engagement and satisfaction.

- Designed and executed scalable data ingestion pipelines using AWS S3 and EC2, ensuring the chatbot can efficiently process and store large volumes of user interactions in real-time, while maintaining high performance under heavy loads.
- Applied advanced Natural Language Processing (NLP) techniques for real-time sentiment analysis and categorization of user feedback, enabling the chatbot to adapt its responses based on user emotions and preferences.
- Developed and fine-tuned NLP models with Hugging Face and Google Dialogflow to enable the chatbot to understand context, handle complex queries, and provide highly accurate, relevant responses to user inputs.
- Engineered decision trees and ensemble methods to predict user intents with higher precision, minimizing response errors, and significantly improving the overall user experience by reducing miscommunication.
- Collaborated with cross-functional teams to integrate the chatbot with existing customer support systems, which streamlined operations, increased response efficiency, and reduced resolution times by 15
- Managed large-scale unstructured conversational data using NoSQL and PL/SQL databases, enabling seamless retrieval and storage for continuous learning and refinement of chatbot models.
- Automated data preprocessing and model retraining workflows by leveraging AWS EC2 instances, ensuring that the chatbot models remain up-to-date with fresh data and capable of adapting to evolving user interactions.
- Developed personalized recommendation algorithms that adapt chatbot interactions based on user history, preferences, and behavior patterns, driving higher user satisfaction and engagement on digital platforms.
- Led A/B testing and validation procedures to ensure the reliability and accuracy of chatbot responses across different business scenarios, continuously improving its performance through rigorous testing.
- Provided strategic insights on AI-driven chatbot development to senior management, offering recommendations on leveraging machine learning and NLP advancements to drive innovation and maintain a competitive edge in customer interaction.

Machine Learning Engineer

Jun 2021 - Dec 2023

Cargill

Irving, TX

- Developed machine learning models using Scikit-learn and Pandas to predict truck delays based on historical route data, weather conditions, and traffic patterns, improving delivery time accuracy and logistics efficiency.
- Engineered real-time data processing pipelines using Spark Streaming and Kafka to analyze live traffic feeds and weather updates, enabling dynamic route adjustments to minimize delays.
- Migrated backend databases from PostgreSQL to Amazon RDS, ensuring scalability and optimized data management, with seamless integration into existing systems.
- Utilized Hopworks for feature engineering and data versioning, improving the efficiency and reproducibility of feature extraction for truck delay prediction models.
- Leveraged Amazon SageMaker to train and deploy predictive models in a fully managed environment, automating the training process and enabling real-time delay predictions.
- Deployed models on Amazon EC2 and containerized applications with Docker, integrating them with Streamlit to provide interactive dashboards for monitoring and visualizing truck delays and predicted arrival times.
- Performed data cleaning and preprocessing using NumPy and Pandas to ensure accurate, high-quality data inputs from multiple sources (GPS, weather APIs, traffic data).
- Automated data workflows with Apache Airflow, streamlining ingestion, processing, and modeling stages for daily predictions and continuous model retraining.
- Developed RESTful APIs with Flask, allowing third-party logistics platforms to access truck delay predictions, improving collaboration and delivery transparency.
- Configured and maintained HDFS environments to store large datasets such as historical traffic data and GPS logs, ensuring efficient and scalable data storage for machine learning applications.
- Integrated CI/CD pipelines with Jenkins, GitLab, and DVC for model versioning and deployment, ensuring seamless updates and scalable infrastructure for machine learning workflows.

- Collaborated with supply chain teams to integrate predictive insights into route planning systems, reducing delays and optimizing resource allocation.
- Built a real-time interactive dashboard in Streamlit, providing stakeholders with dynamic visualizations of truck locations, predicted delays, and performance metrics for operational decision-making.
- Documented all project phases and outcomes, ensuring the entire machine learning pipeline from data collection to deployment was transparent and reproducible for future development.
- Leveraged Kubernetes for container orchestration, ensuring scalability, reliability, and efficient deployment of machine learning applications in production.
- Utilized ML frameworks such as TensorFlow, PyTorch, and model management tools like MLflow and Kubeflow for streamlined model training and deployment.
- Automated infrastructure provisioning and configuration using tools like Terraform, Ansible, and CloudFormation, ensuring robust and repeatable environment setups.
- Maintained cloud infrastructure on AWS, utilizing services such as SageMaker, RDS, EC2, and S3, while integrating predictive insights with serverless architecture to optimize cost and performance.

Data Scientist

Jan 2017 - Jun 2021

MU-SIGMA

Bangalore, India

- Analyzed clinical trial data using Python and Pandas to identify patterns and predict trial outcomes, enhancing the effectiveness and efficiency of research initiatives.
- Developed machine learning models with Scikit-learn and TensorFlow to predict patient responses to treatments, improving personalized medicine strategies.
- Engineered robust data pipelines using Spark Core, Spark SQL, and Spark Streaming to handle real-time data processing, enhancing data accessibility and analysis.
- Utilized Amazon EMR, EC2, and S3 for scalable cloud-based data analysis, ensuring efficient management of vast datasets typical in pharmaceutical research.
- Implemented Airflow to automate and schedule complex data workflows, increasing the productivity and reliability of data operations.
- Designed predictive maintenance models for laboratory equipment using machine learning, reducing downtime and optimizing operational efficiency.
- Conducted text analytics and natural language processing on medical literature and patent databases using Python, extracting valuable insights for drug development.
- Created data visualization tools using Matplotlib and Pandas to present research findings to stakeholders, facilitating data-driven decision-making.
- Managed MongoDB and Hive databases to support efficient data storage and retrieval, optimizing data architecture for high-performance needs.
- Developed RESTful APIs using Flask to integrate machine learning models into healthcare applications, enhancing user access to predictive analytics.
- Configured and maintained HDFS for robust data storage solutions, ensuring data integrity and security in compliance with industry regulations.
- Utilized GitHub for version control, maintaining a collaborative and efficient development environment across the data science team.
- Enhanced supply chain management by implementing machine learning algorithms that predict demand and optimize inventory levels.
- Applied Spark Streaming to monitor real-time data from clinical operations, enabling immediate insights and responses to operational anomalies.
- Collaborated with cross-functional teams to integrate data-driven insights into pharmaceutical and healthcare products, ensuring alignment with business goals.

- Automated the extraction, transformation, and loading (ETL) of data using Airflow, streamlining the data preparation process for advanced analytics.
- Led initiatives to analyze and improve data quality across various data sources using Python and Scikit-learn, ensuring the reliability of analytical models.
- Documented all data science processes and findings in a clear and comprehensive manner, maintaining detailed records to support regulatory submissions and internal audits.

Data Analyst

Fractal Analytics

Jan 2014 - Jan 2017

Hyderabad, India

- Utilized advanced SQL techniques for data manipulation and optimization, including stored procedures, functions, and complex joins, to support in-depth analysis and reporting.
- Leveraged BeautifulSoup and Requests libraries in Python for web scraping, extracting valuable data from web pages.
- Used SSIS for ETL processes, designing packages to automate data extraction, transformation, and loading, improving data accuracy and efficiency.
- Developed Python scripts with PyPDF2 and pdfplumber for automated PDF data extraction, integrating these scripts into SSIS packages to enhance data workflows.
- Implemented advanced SQL indexing techniques to significantly enhance query performance.
- Designed and implemented Power BI solutions to automate reporting, reducing manual effort and providing real-time insights.
- Integrated Power BI with various data sources for seamless connectivity and accessibility.
- Ensured data governance by maintaining accuracy, consistency, and security across all data lifecycle stages.
- Maintained Power BI dashboards with advanced interactive features for dynamic data exploration and detailed reporting.
- Leveraged AWS services such as S3 for data storage and EC2 for processing, deploying scalable and cost-effective analytical solutions.
- Conducted data wrangling processes to clean and transform raw data, ensuring quality and consistency throughout the analysis.